

Trainings ▾

General Information

Petroleum diesel contains chemical compounds known as paraffins and aromatics. Paraffinic fuels are predominantly paraffins and are derived from a variety of sources. The most common sources are natural gas, coal, plant oils, and animal fats. Natural gas fuels are often referred to as gas- to-liquid (GTL). Fuels from plant oils and fats are often referred to as renewable diesel (RD) or hydrogenated vegetable oil (HVO). Renewable diesels are **not** the same as biodiesel or fatty acid methyl ester (FAME) which are chemically different and have different properties. Unique characteristics of paraffin-based fuel include reduced density, limited sulfur/aromatics content, higher hydrogen to carbon (H/C) ratio, and increased cetane. Generally, paraffinic fuels have a density about 7 percent lower than petroleum diesels and can result up to in about 5 percent lower energy content by volume, resulting in reduced fuel economy and possible reduction in engine power.

Cummins Inc. testing indicates that paraffinic fuels generally result in lower smoke emissions, reduced power output, and decreased fuel economy. The European specification for paraffinic fuels is defined in EN 15940. Paraffinic fuels **must** meet the requirements of either Table 8 of this section or EN 15940. The requirements of EN 15940 **must** be achieved. The minimum requirement is to meet Section 1: Diesel Fuel (00-375-001), excluding the density requirement in Table 1, Cummins Required Fuel Properties. These fuels are approved for use in the engines listed in Table 9: Products and Applications Approved for up to 100% Paraffinic Fuel Blends.

Note : For all other Cummins® products that are not listed in Table 9 are not approved to 100 percent blends, the Paraffinic component fuel is allowed in any blend proportions provided that the final blended diesel fuel complies with EN 590 or the requirements listed in Table 1 in this bulletin for all Cummins® engines.

The user is responsible to verify/obtain the proper local, regional, or national exemptions required for the use of paraffinic fuels in any emissions regulated Cummins® engine.

Paraffinic fuels like conventional diesel fuel can be affected by environmental factors such as cold temperatures that can inhibit proper operation of Cummins engines®. The user is responsible for procuring paraffinic fuel that conforms to the Cloud Point requirements in Table 8 and based temperature factors in the region of use.

Warranty and the Use of Paraffinic Fuel in Cummins® Engines

Cummins® Engine Warranty covers malfunctions that are a result of defects in material or factory workmanship. Engine damage, service issues, and/or performance issues determined by Cummins® to be caused by using paraffinic fuel **not** meeting the specifications outlined in this document are **not** considered to be defects in material or workmanship and are **not** covered under Cummins® engine warranty.

Table 8, Summary of EN 15940 Requirements for Paraffinic Fuels

Performance Characteristics	Unit	Limits Class A		Limits Class B		Test Procedures
		Min	Max	Min	Max	
Cetane number		70.0	-	51.0	-	EN 15195 Pr EN16906 EN ISO 5165
Density at 15°C	kg/m ³	765.0	800.0	780.0	810.0	EN ISO 3675 EN ISO 12185
Flashpoint	°C	55.0	-	55.0	-	EN ISO 2719
Viscosity at 40°C	mm ² /s	2.000	4.500	2.000	4.500	EN ISO 3104
Distillation % (V/V) recovered at 250°C % (V/V) recovered at 350°C 95% (V/V) recovered at	% (V/V) % (V/V) °C	- 85 -	65 - 360	- 85 -	65 - 360	EN ISO 3405 EN ISO 3924
Lubricity, wear scar diameter (wsd) at 60°C	µm	-	460	-	460	EN ISO 12156-1
FAME content	%(V/V)	NOTE A				EN 14078
Manganese content	mg/l	-	2.0	-	2.0	EN 16136
Total aromatic contents	%(m/m)	-	1.1	-	1.1	EN 15940 Annex C

Sulfur content	mg/kg	-	5.0	-	5.0	EN ISO 20846 EN ISO 20884
Carbon residue (on 10% distillation residue)	%(m/m)		0.30		0.30	EN ISO 10370
Ash content	%(m/m)	-	0.010	-	0.010	EN ISO 6245
Water content	mg/kg	-	200	-	200	EN ISO 12937
Total contamination	mg/kg	-	24	-	24	EN 12662
Copper strip corrosion (3 h at 50°C)	rating	Class 1		Class 1		EN ISO 2160
Cloud Point	°C (°F)	Must meet lowest expected ambient temperature		Must meet lowest expected ambient temperature		ASTM D2500
Oxidation stability	g/m ³	-	25	-	25	EN ISO 12205
	h	20.0	-	20.0	-	EN 15751

A. Max 7 percent in EU; Max 5 percent in the US.

Table 9, Product and Applications Approved for up to 100% Blend of Paraffinic Fuels, A

Product Series	Displacement (L)	Products	Notes
B/D	2.8, 3.3, 3.9, 4.5, 5.9, 6.7	All engines	
F	2.8, 3.8, 4.5	All engines	
C/L	8.3, 8.9	All engines	
M	11	All engines	
G/X	12	All engines	B, C
X/QSX	15	All engines except models shown in Note D	C, D, E
ISX/QSX 12	12	Not permitted	

Table 9, Product and Applications Approved for up to 100% Blend of Paraffinic Fuels, A

Product Series	Displacement (L)	Products	Notes
K, QSK, QST	19, 23, 30, 38, 45, 50, 60, 78, 95	All engines	
M	14	All engines produced by Dongfeng - Cummins® Engine Company	
Z, M	14, 15	All engines produced by Dongfeng - Cummins® Engine Company	
QSZ	13	All engines produced by Dongfeng - Cummins® Engine Company	
N	14	All engines	

Notes:

- A. Where applicable, both On- and Off-highway applications are considered. Please see additional notes below for further definition on which applications are approved.
- B. X12 is the on-highway application engine, and QSG12/X12 CM2350 G110 is the off-highway application engine.
- C. While running with Paraffinic Fuel, check or reset valve lash at half the specified service interval. Refer to owner's manual.
- D. The ISX15 CM2350 X101 is **not** compatible with paraffinic fuel.
- E. QSX15 CM2350 X105 **must** be built or have an ECM calibration code released after June 2022. QSX15 CM2350 X106 **must** be built or have an ECM calibration code released after January 23, 2023. X15 CM2350 X130C **must** be built or have an ECM calibration code released after November 9, 2022. X15 CM2350 X140C **must** be built or have an ECM calibration code released after December 19, 2022.